MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
6000 VILLAGE:GUEST HOUSE
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of Guest House in Village Area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Guest House only.

4. DESIGN METHODOLOGY & ASSUMPTIONS

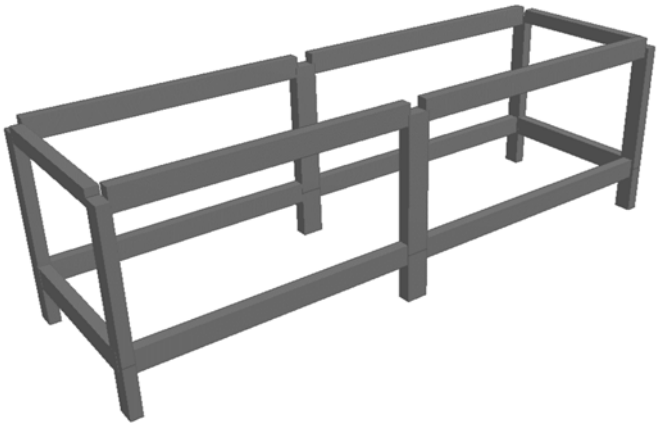
This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HSC (Hollow Core Slab)

The Superstructure for this Accommodation Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ Columns & Beams) has been performed meeting design requirements & also keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base which are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.

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STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007-Structural Loading & Forces.

[2] SBC-303-2007-Structural Soil & Foundations.

[3] SBC-304-2007-Structural Concrete Structures.

[4] SASO SSA 2 – Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 – Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 Building Code Requirements.

[8] PCI –MNL-120-04 –Design Handbook, 6th edition.

4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-CAL-1008-A02.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

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Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel

5. DESIGN LOADS

5.1. Dead Load

5.1.1. Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²

- Hollow Core Slab (PCI 6th Edition)

(HCS-150 Topped) = **2.52 KN/m2**

Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt-KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m² for Finishes Characteristic Service Loads Kn/m²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

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0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/m²**

Total Load = **5.82 kN/m²**

UDL on Main Beams (LB1) = 5.82 x 3.95/2 =**11.50 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+(0.05+0.2)/2x20)*0.6*0.6/2 = **1 knm/m**

5.1.2. Ground Floor Walls

- 250 thk. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x 2. 5m clear height = **10.0 kN/m**

5.2. Live Load

5.2.1. Roof Live Loads

Live loads were taken from SEI/ASCE 7-05, Table 4-1.

Ordinary Flat(Non-Accessible) = **0.50 kN/m²**

Udl Due to Live Load on the Ledger Beam (LB1) = 0.50X4.2/2 = **1.05 kN/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

5.3. Wind Load

Wind loads were automatically generated in STAAD as per ASCE 7-05. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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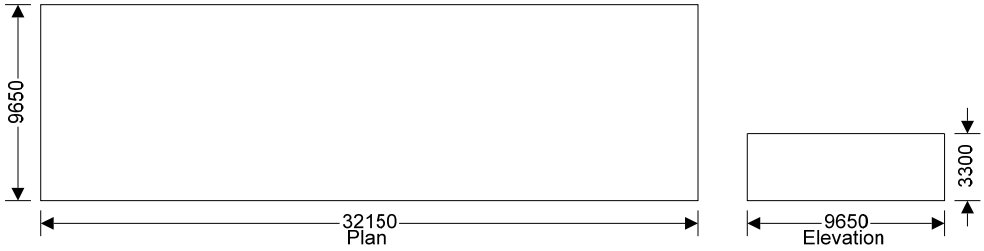
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5.4. Wind Load (Analytical Verification)

WIND LOADING (SBC-301)

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 10300 mm
Width of building;	d = 4200 mm
Height to eaves;	H = 3300 mm
Mean height;	h = 3300 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff;	K_d = 0.85(As per SBC-301, Table 6.4-1)
Exposure category ;	D(As per SBC-301, Table 6.4.2.3)
Enclosure classification ;	Partially enclosed buildings(As per SBC-301, Section-6.2)
Internal pressure coef +ve;	GC_{pi_p} = 0.55(As per SBC-301, Figure-7.2-1)
Internal pressure coef –ve;	GC_{pi_n} = -0.55(As per SBC-301, Figure-7.2-1)
Gust effect factor;	G_f = 0.85(As per SBC-301, Section-7.2.7.1)
Importance Factor;	1(As per SBC-301, Section-9.5)

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Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2 \times I$

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	K_z (SBC-301,Table 7.2-2)	q_z (kN/m^2)
3300	1.03	1.29
3301	1.03	1.29

Peak velocity pressure for internal pressure

Peak velocity pressure – internal (as roof press.); q_i = 1.289 kN/m^2

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	34.00	5.70
B	3.30	-0.50	1.29	-1.26	33.99	-42.70
C	3.30	-0.70	1.29	-1.48	13.86	-20.45
D	3.30	-0.70	1.29	-1.48	13.86	-20.45

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = b \times H$ = 33.99 m^2

Projected vertical area of roof; $A_{vert_r_0}$ = 0.00 m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0}$ = 26.2 kN

Leeward net force; $F_l = F_{w,wB}$ = -42.7 kN

Windward net force; $F_w = F_{w,wA}$ = 5.7 kN

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Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 48.4Kn$

Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	34	53.89
B	3.30	-0.50	1.29	0.16	33.99	5.47
C	3.30	-0.70	1.29	-0.06	13.86	-0.80
D	3.30	-0.70	1.29	-0.06	13.86	-0.80

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = b \times H = 33.99\ m^2$

Projected vertical area of roof; $A_{vert_r_0} = 0.00\ m^2$

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} = 26.2\ kN$

Leeward net force; $F_l = F_{w,wB} = 5.5\ kN$

Windward net force; $F_w = F_{w,wA} = 53.9\ kN$

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min}) = 48.4Kn$

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	0.17	13.86	2.32
B	3.30	-0.23	1.29	-1.01	13.86	-14.03
C	3.30	-0.70	1.29	-1.48	33.99	-50.15
D	3.30	-0.70	1.29	-1.48	33.99	-50.15

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = d \times H = 13.86\ m^2$

Projected vertical area of roof; $A_{vert_r_0} = 0.00\ m^2$

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0} = 10.7\ kN$

Leeward net force; $F_l = F_{w,wB} = -14.0\ kN$

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Windward net force; $F_w = F_{w,wA}$ =2.30 kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min})$ = 16.4Kn

Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A	3.30	0.80	1.29	1.58	13.86	21.97
B	3.30	-0.23	1.29	0.45	13.86	5.61
C	3.30	-0.70	1.29	-0.06	33.99	-1.97
D	3.30	-0.70	1.29	-0.06	33.99	-1.97

Overall loading

Projected vertical plan area of wall; $A_{vert_w_0} = d \times H$ = 13.86 m^2

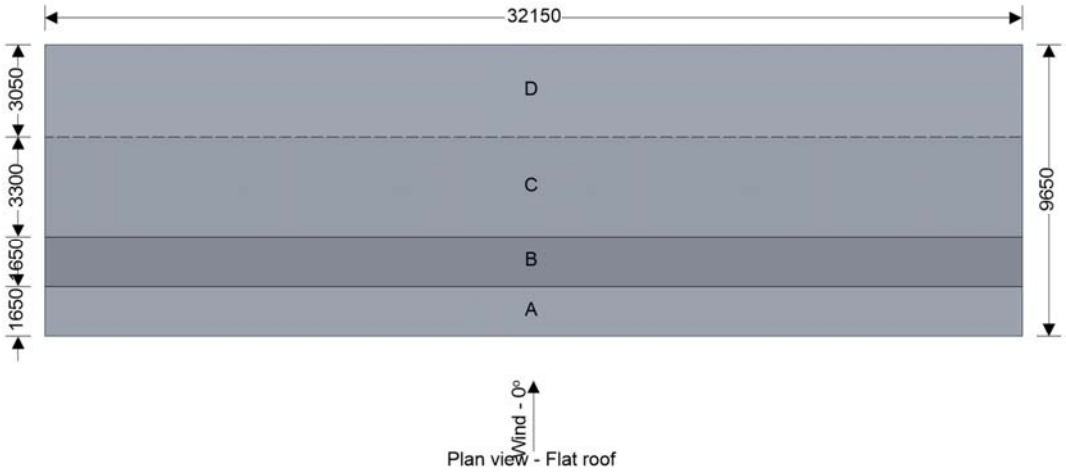
Projected vertical area of roof; $A_{vert_r_0}$ = 0.00 m^2

Minimum overall horizontal loading; $F_{w,total_min} = p_{min_w} \times A_{vert_w_0} + p_{min_r} \times A_{vert_r_0}$ = 10.7 kN

Leeward net force; $F_l = F_{w,wB}$ = 5.60 kN

Windward net force; $F_w = F_{w,wA}$ =22.0 kN

Overall horizontal loading; $F_{w,total} = \max(F_w - F_l + F_{w,h}, F_{w,total_min})$ = 16.4Kn

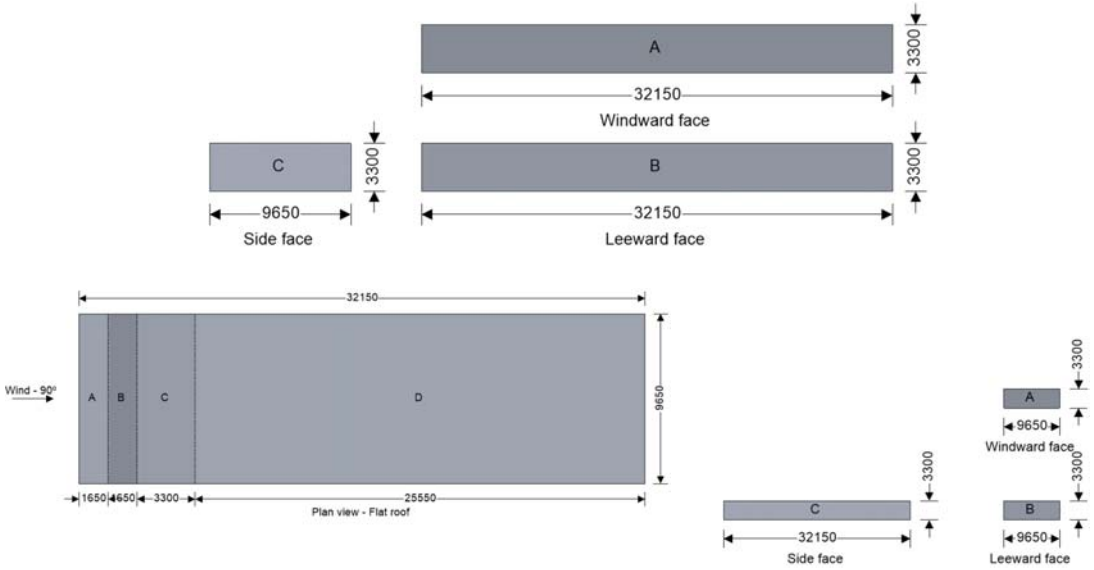


Plan view - Flat roof

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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

Seismic Load

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.10 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@S1

S_{DS} Damped Design Spectral Response Acceleration@Ss

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Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.5. Thermal/Temperature Loads (Self-restraining and Thermal Expansion and Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

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Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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8. DEFLECTIONS

8.1. Maximum Allowable Deflection

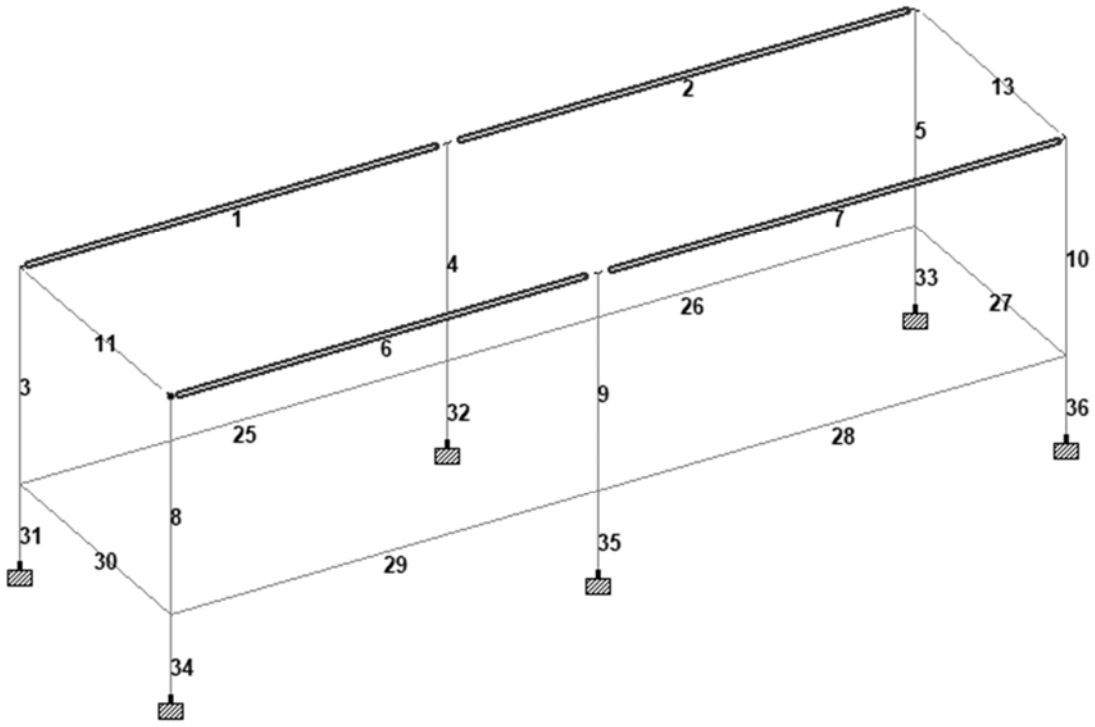
L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support

9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam (LB1)



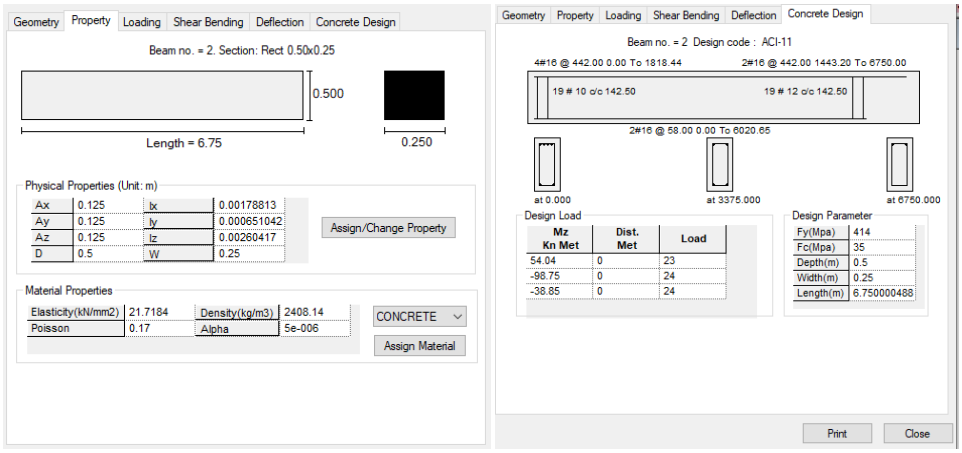
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All / Summary / Envelope

	Beam	L/C	Node	Fx kN	Fy kN	Fz kN	Mx kNm	My kNm	Mz kNm
Max Fx	35	23 ULS: (1.4	27	**308.163**	-0.483	5.108	0.055	1.246	-1.507
Min Fx	13	31 ULS: (0.9	12	**-21.955**	-3.301	0.000	-1.599	-3.923	4.098
Max Fy	7	23 ULS: (1.4	11	22.175	**78.842**	0.225	-3.119	-0.874	97.329
Min Fy	6	23 ULS: (1.4	11	16.872	**-73.306**	-0.272	2.485	-0.952	86.726
Max Fz	36	28 ULS: (1.2	9	181.105	23.297	**46.119**	-1.822	-10.778	18.921
Min Fz	33	22 ULS: (1.4	3	169.559	26.679	**-17.995**	0.264	15.518	21.906
Max Mx	1	27 ULS: (1.2	4	18.903	44.653	3.840	**19.004**	-10.140	18.112
Min Mx	6	27 ULS: (1.2	10	18.902	44.653	-3.841	**-19.004**	10.141	18.111
Max My	36	28 ULS: (1.2	28	184.506	23.297	46.119	-1.822	**44.565**	-9.035
Min My	5	26 ULS: (1.2	6	40.981	17.569	-14.922	-2.274	**-20.969**	-34.963
Max Mz	2	23 ULS: (1.4	5	22.173	78.842	-0.226	3.119	0.874	**97.329**
Min Mz	32	32 ULS: (0.9	24	196.801	51.199	3.236	0.877	0.570	**-95.081**



Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = **98 Knm (Hogging)**

Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-16/2=442mm, Fc'=35 Mpa & ϕ =0.9

Ast.req = **619 mm²**

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Ast.min=
(As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **54 Knm (Sagging)**

Hence Provide 3-20 dia-Full Straight Bars Top & 3-16 dia-Full Straight Bottom considering

Ast.pro**>>>>>>(Ast.min, Ast.req)**

9.1.1. Staad Output For LB1

```
        B E A M  N O.     2 D E S I G N  R E S U L T S - SHEAR

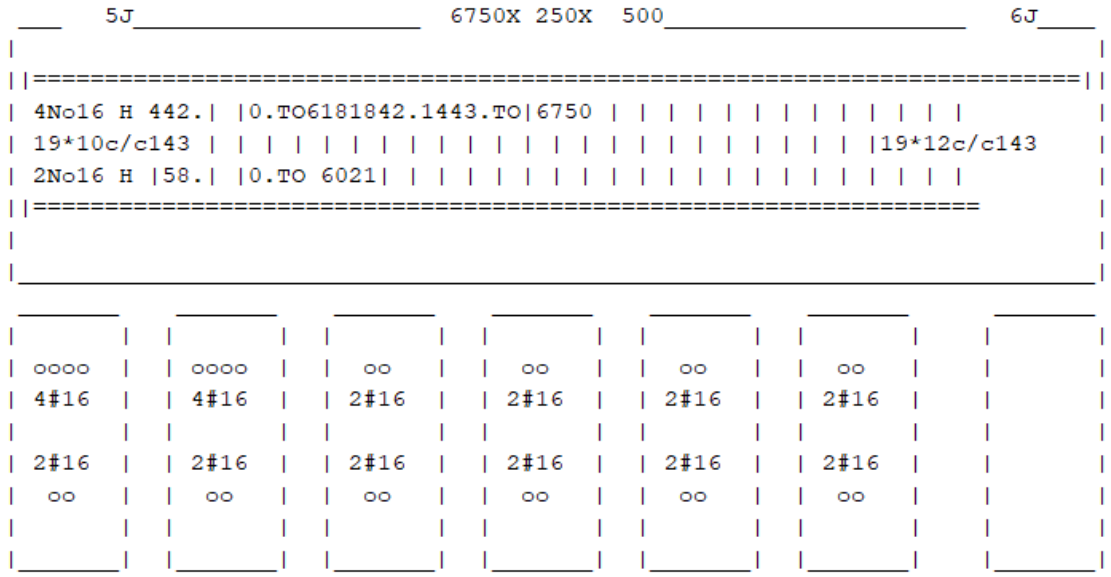

AT START SUPPORT - Vu=   53.99 KNS  Vc=  108.63 KNS  Vs=    0.00 KNS
Tu=   14.69 KN-MET  Tc=    3.9 KN-MET  Ts=   19.6 KN-MET  LOAD    26
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2495. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.84 SQ.CM.

AT END   SUPPORT - Vu=   39.19 KNS  Vc=  175.20 KNS  Vs=    0.00 KNS
Tu=   19.13 KN-MET  Tc=    4.0 KN-MET  Ts=   25.5 KN-MET  LOAD    27
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 12 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 2495. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   6.30 SQ.CM.
```

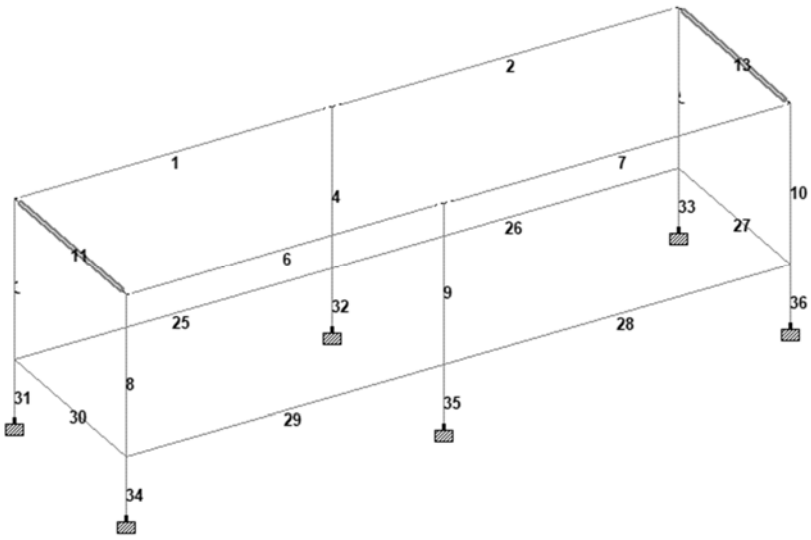
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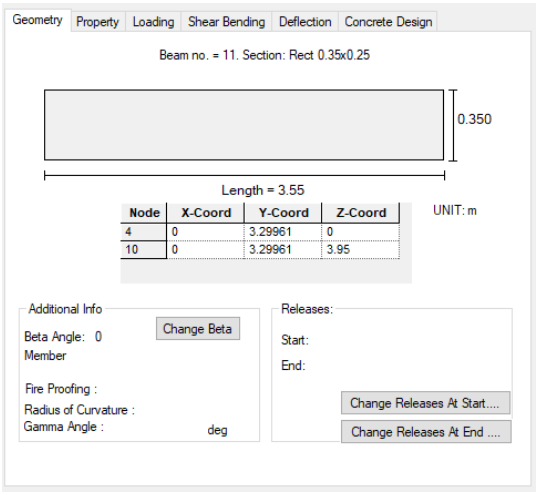
9.2. Design of Beam-(B1)

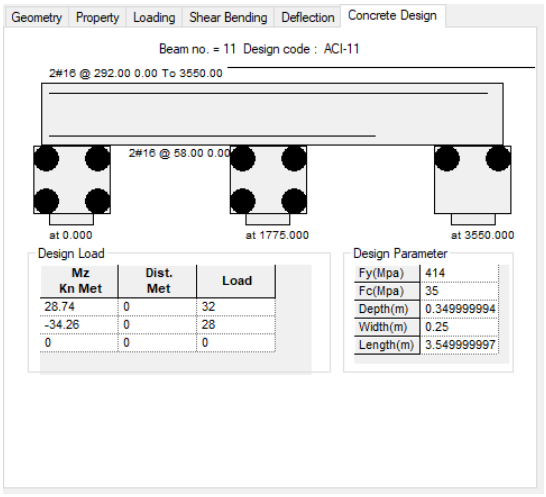


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Design of Beam (B1)

Hence Provide 2-16 dia-Full Straight Bars Top & Bottom.

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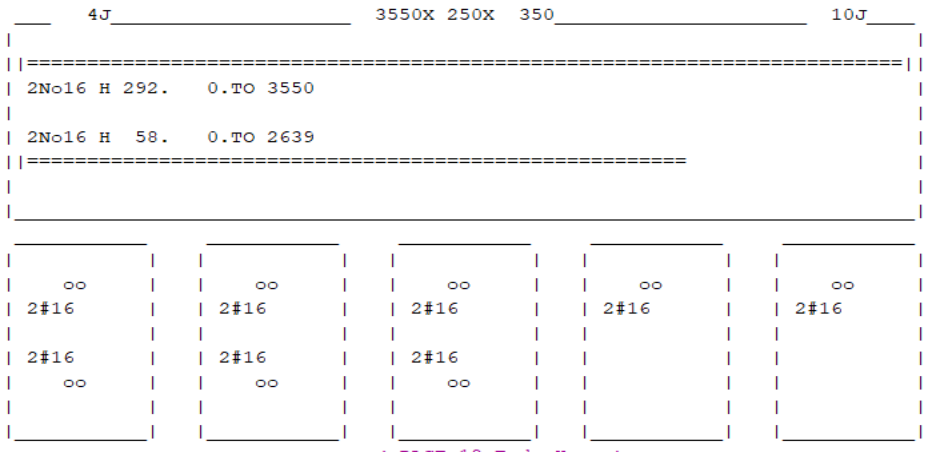
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9.2.1. Staad Output For B1

```
                B E A M   N O.    11 D E S I G N   R E S U L T S - SHEAR


AT START SUPPORT - Vu=     2.47 KNS   Vc=    71.55 KNS   Vs=     0.00 KNS
Tu=     1.19 KN-MET   Tc=     2.4 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=     2.47 KNS   Vc=    71.54 KNS   Vs=     0.00 KNS
Tu=     1.20 KN-MET   Tc=     2.4 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                    STIRRUPS ARE NOT REQUIRED.
```



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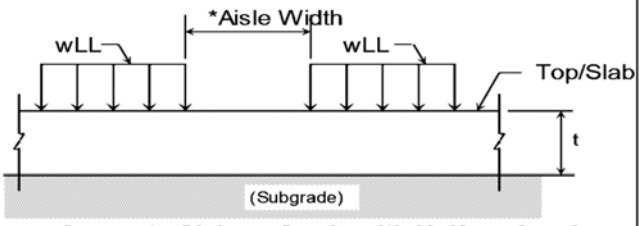
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10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS
For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Allow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

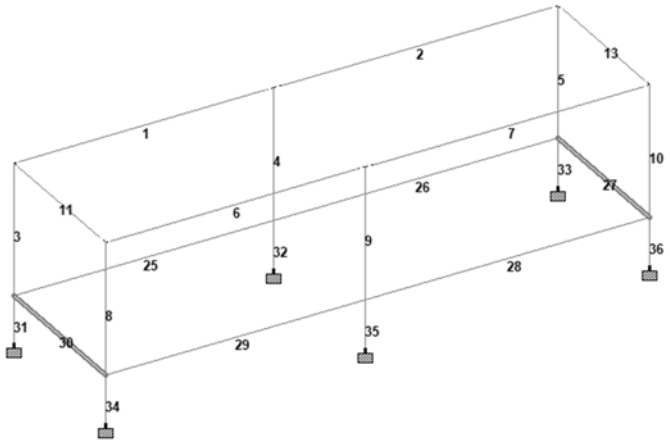
Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

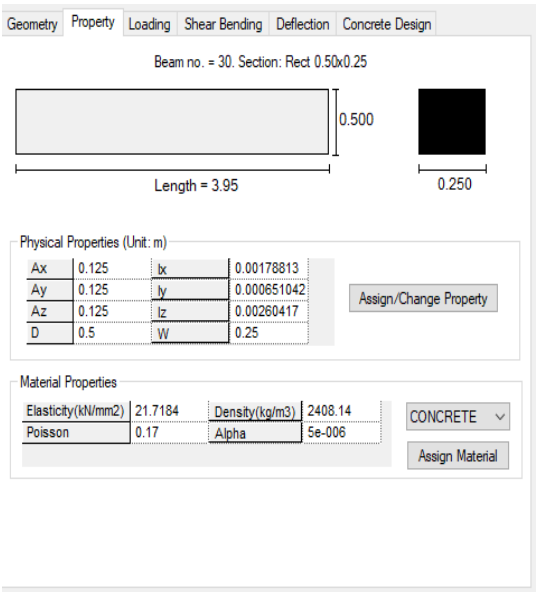
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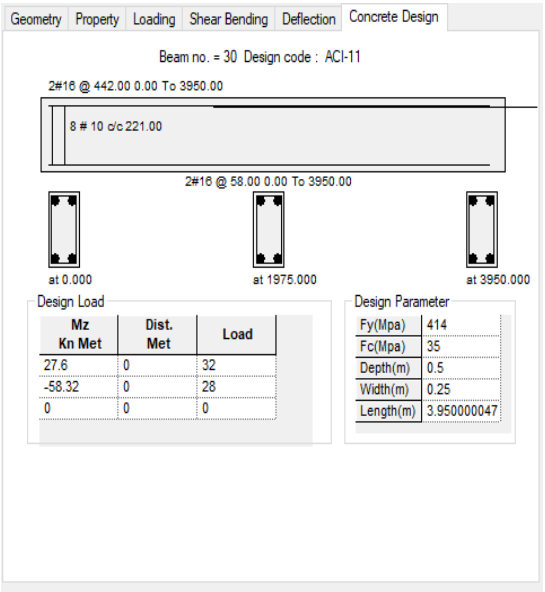
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11. DESIGN OF BEAM-(GB1)







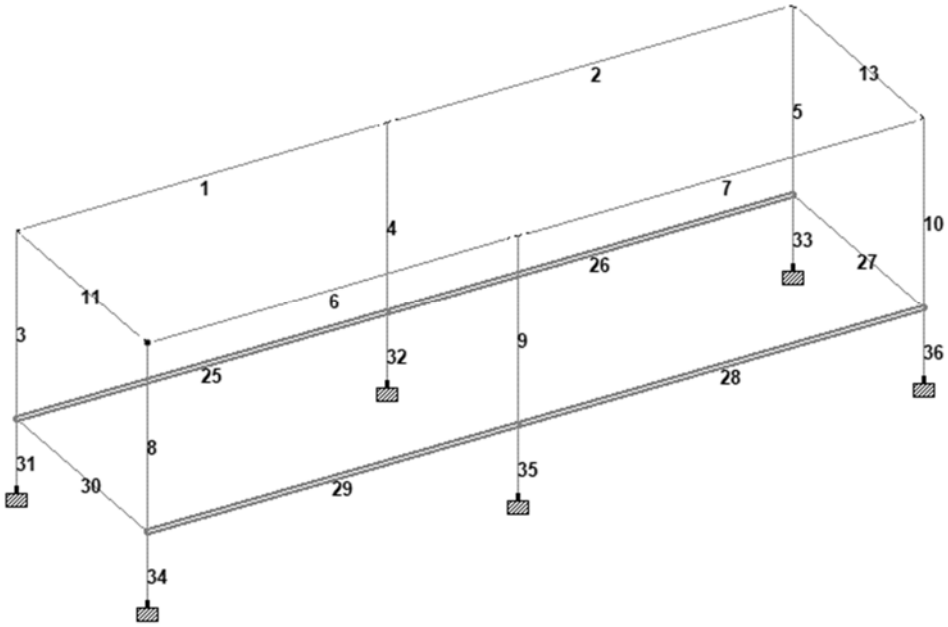
However Provide minimum Reinf. 3-16 dia-Full Straight Bars Top & Bottom

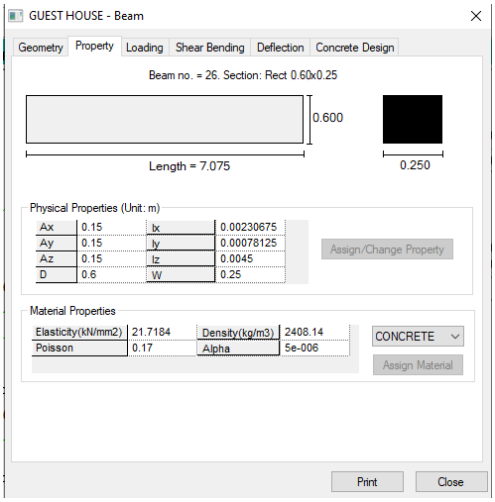
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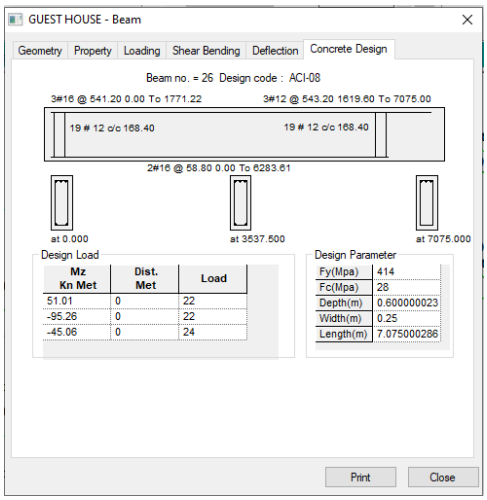
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12. DESIGN OF BEAM-(GB2)







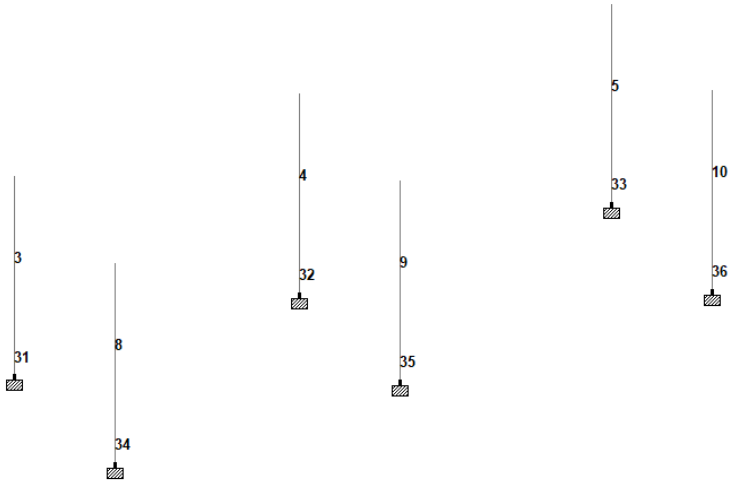
However Provide minimum Reinf. 3-16 dia-Full Straight Bars Top & Bottom

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13. CONCRETE COLUMN DESIGN



Column numbers

13.1. Staad Output For Column C1

```
       ACI 318-11    COLUMN  NO.      3    DESIGN RESULTS
       ================================================


 FY - 413.7   FC -   35.0 MPA,   RECT SIZE - 400.0 X 250.0 MMS, TIED
          AREA OF STEEL REQUIRED =   1214.0   SQ. MM

 BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
 ------------------------------------------------------------

   8 - 16 MM                 1.609        26       END      0.650
 (PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
 TIE BAR NUMBER    10 SPACING 250.00 MM
```


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13.2. Staad Output For Column C2

```
      ACI 318-11   COLUMN  NO.     4   DESIGN RESULTS
      ================================================


FY - 413.7  FC -  35.0 MPA,  RECT SIZE - 400.0 X 250.0 MMS, TIED
        AREA OF STEEL REQUIRED =  1058.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  8 - 16 MM               1.609       32      STA     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM
```

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14. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties		
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r"	24	kN/m^3
Clear Span	0.6	met	Strength of concrete "f_c"	35	N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420	N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20	mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9	
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75	
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05	N/mm^2
Loading			Modulus of Elasticity of concrete	27806	N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19	
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**		
Wall load	0	kN/m^2	Top layer Nos.	6 Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0 Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"	468	mm^2
Total Ultimate Load "Uu_{tot}"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"	0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"db' = 10 mm		

Calculations

Shear force Vu = U*le	**8.096**	**kN**
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**

Check for Section capacity

Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm					
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	**>**	**3.51**	**kNm**	(Mu)

Shear requirements

Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	**>**	**8.096**	**kN**	(Vu)

Serviceability requirements

d' = c+d_b/2	26	mm
B = b/(n*$A_{s\ t}$)	0.297	
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0	
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'	3.915E+07	mm^4
Calculate Gross moment of inertia		
Area of slab A = t*b	2E+05	mm^2
Depth of neutral axis "y"	75	mm
Distance of top fiber from N.A "y_t"	75	mm
Gross moment of inertia "I_g"	2.813E+08	mm^4
Allowable stress fr = 0.62√f'c	3.67	mm^2
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm
Moment due to live load "Mll"	0.32	kNm
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm

Deflection due to dead + live loads

R(dl+ll) = M_{cr}/Ms	1.000		
I_e(dl+ll) = $R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
δ = K*5/48*Ms*le^2/(Ec*I_e)	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411	
I_e(dl) = $R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4
δ = K*5/48*Md*le^2/(Ec*I_e)	0.00	mm

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load = per ACI 318 Tab 9.5b **I_e/360** **2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
M_{sus} = M_{dl}+SL*M_{ll}	2.70	kNm
R(sus) = M_{cr}/Msus	5.091	
I_e(sus) = $R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
δ = K*5/48*Msus*le^2/(Ec*I_e)	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = I_e/240 **3.333 mm > 0.079 mm O.K**

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15. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-CAL-1008 for Details)

TABLE 10: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Area	Borehole No.	Proposed Structures	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement	Vertical Modulus of Subgrade Reaction (Ks)	Horizontal Modulus of Subgrade Reaction (Kh)
						Settlement	Shear Failure				
				(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)	($kN/m^2/m$)	(kN/m^2)
Village Area	BH-21	Village	Isolated Square	1.20	1.0 x 1.0	1000	350	350	25	42,000	32,200
				1.50		1100	500	500	25	60,000	38,900
				1.20	2.0 x 2.0	700	580	580	25	69,600	37,500
				1.50		850	750	750	25	90,000	44,200
				1.20	3.0 x 3.0	600	750	600	25	72,000	42,800
				1.50		700	930	700	25	84,000	49,500
				1.20	4.0 x 4.0	550	910	550	25	66,000	48,100
				1.50		625	1090	625	25	75,000	54,800
				1.20	5.0 x 5.0	475	1060	475	25	57,000	53,400
				1.50		550	1240	550	25	66,000	60,200

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.)	BUILDING PLANS	MD-216-6000-EG-AR-GAD-3087
2.)	OVERALL LAYOUT PLAN	MD-216-6000-EG-GAD-1535
3.)	LAYOUT & DETAILS OF FOUNDATION, COLUMN & GRADE BEAM	MD-216-6000-EG-CV-DED-1487
4.)	LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-6000-EG-CV-CAL-1488
5.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
6.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001

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ATTACHMENT 1 STAAD INPUT DATA

```
ST STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 6.475 0 0; 3 13.55 0 0; 4 0 3.29961 0; 5 6.475 3.3 0;
6 13.55 3.29961 0; 7 0 0 3.95; 8 6.475 0 3.95; 9 13.55 0 3.95;
10 0 3.29961 3.95; 11 6.475 3.3 3.95; 12 13.55 3.3 3.95; 23 0 -1.2 0;
24 6.475 -1.2 0; 25 13.55 -1.2 0; 26 0 -1.2 3.95; 27 6.475 -1.2 3.95;
28 13.55 -1.2 3.95;
MEMBER INCIDENCES
1 4 5; 2 5 6; 3 1 4; 4 2 5; 5 3 6; 6 10 11; 7 11 12; 8 7 10; 9 8 11; 10 9 12;
11 4 10; 13 6 12; 25 1 2; 26 2 3; 27 3 9; 28 9 8; 29 8 7; 30 7 1; 31 1 23;
32 2 24; 33 3 25; 34 7 26; 35 8 27; 36 9 28;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
3 TO 5 8 TO 10 31 TO 36 PRIS YD 0.25 ZD 0.4
1 2 6 7 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
25 TO 30 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
11 13 PRIS YD 0.35 ZD 0.25
CONSTANTS
BETA 90 MEMB 4 9 32 35
MATERIAL CONCRETE ALL
SUPPORTS
23 TO 28 FIXED
*******************************************************************************
MEMBER CRACKED
3 TO 5 8 TO 10 31 TO 36 REDUCTION RIY 0.700000 RIZ 0.700000
1 2 6 7 11 13 27 30 REDUCTION RIY 0.350000 RIZ 0.350000
```


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```
MEMBER OFFSET
2 7 START 0.2 0 0
2 7 END -0.125 0 0
1 6 START 0.125 0 0
1 6 END -0.2 0 0
11 13 START 0 0 0.2
11 13 END 0 0 -0.2
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 2
SELFWEIGHT 1
MEMBER WEIGHT
27 30 UNI 10
MEMBER WEIGHT
1 2 6 7 UNI 8.9
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 14.000 M 5.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.900 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.6486 1.6486 HEIG  0  4.572
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 14.000 M 5.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.900 -0.200 0.550
!> END GENERATED DATA BLOCK
INT -0.947728 -0.947728 HEIG  0  4
TYPE 3 SIDEWARD X
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 15.000 M 5.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.900 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.53179 -1.53179 HEIG  0  4
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 5.000 M 15.000 M 2.000 0.010 0 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
```


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0 0 0 0.886 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.63434 1.63434 HEIG 0 4.572
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 5.000 M 15.000 M 2.000 0.010 1 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.886 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.28925 -1.28925 HEIG 0 4
TYPE 6 SIDEWARD Z
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 4.000 M 5.000 M 15.000 M 2.000 0.010 2 -
0 0 0 0 1.030 1.000 1.000 0.850 0 -
0 0 0 0.886 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.51931 -1.51931 HEIG 0 4
LOAD 1 LOADTYPE Seismic TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
25 TO 30 UNI GY -10
MEMBER LOAD
1 2 6 7 UNI GY -11.5
11 UMOM GZ 1
13 UMOM GZ -1
1 2 UMOM GX -1
6 7 UMOM GX 1
LOAD 4 LOADTYPE Live TITLE LL
MEMBER LOAD
1 2 6 7 UNI GY -2.1
11 UMOM GZ 0.2
13 UMOM GZ -0.2
1 2 UMOM GX -0.2
6 7 UMOM GX 0.2

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LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 4
WIND LOAD -X 1 TYPE 2 YR 0 4
WIND LOAD -Z 1 TYPE 3 YR 0 4
WIND LOAD -Z -1 TYPE 3 YR 0 4
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 4
WIND LOAD -Z 1 TYPE 5 YR 0 4
WIND LOAD -X 1 TYPE 6 YR 0 4
WIND LOAD -X -1 TYPE 6 YR 0 4
LOAD 7 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
1 TO 11 13 TEMP 25
LOAD COMB 8 SLS: (DL)
3 1.0
LOAD COMB 9 SLS: (DL+T)
3 1.0 7 1.0
LOAD COMB 10 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 11 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 12 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 13 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 14 SLS: (D+0.75WX+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 15 SLS: (D+0.75WZ+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 16 SLS: (0.6D+WX)
3 0.6 5 1.0
LOAD COMB 17 SLS: (0.6D+WZ)
3 0.6 6 1.0
LOAD COMB 18 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 19 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 20 SLS: (D+WX)
3 1.0 5 1.0
LOAD COMB 21 SLS: (D+WZ)
3 1.0 6 1.0
LOAD COMB 22 ULS: (1.4 DL)
3 1.4
LOAD COMB 23 ULS: (1.4D+1.4T+0.5LR)

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3 1.4 4 0.5 7 1.4
LOAD COMB 24 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5 7 1.4
LOAD COMB 25 ULS: (1.2D+0.80WX+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 26 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 27 ULS: (1.2D+1.6WX+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 29 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 30 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 31 ULS: (0.90D+1.6WX)
3 0.9 5 1.6
LOAD COMB 32 ULS: (0.90D+1.6WZ)
3 0.9 6 1.6
LOAD COMB 33 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 34 ULS: (0.90D+EX)
3 0.9 2 1.0
PERFORM ANALYSIS
START CONCRETE DESIGN
CODE ACI
FC 35000 MEMB 1 TO 11 13 27 30 TO 36
EFACE 0.44 MEMB 1 2 6 7
SFACE 0.44 MEMB 1 2 6 7
EFACE 0.29 MEMB 11 13
SFACE 0.29 MEMB 11 13
CLB 0.04 MEMB 1 TO 11 13 27 30 TO 36
CLS 0.04 MEMB 1 TO 11 13 27 30 TO 36
CLT 0.04 MEMB 1 TO 11 13 27 30 TO 36
MAXMAIN 16 MEMB 1 TO 11 13 27 30 TO 36
MINMAIN 16 MEMB 1 TO 11 13 27 30 TO 36
MINMAIN 20 MEMB 32
MAXMAIN 20 MEMB 32
MINSEC 10 MEMB 1 TO 11 13 27 30 TO 36
DESIGN BEAM 1 2 6 7 11 13 25 TO 30
DESIGN COLUMN 3 TO 5 8 TO 10 31 TO 36
END CONCRETE DESIGN
FINISH

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ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

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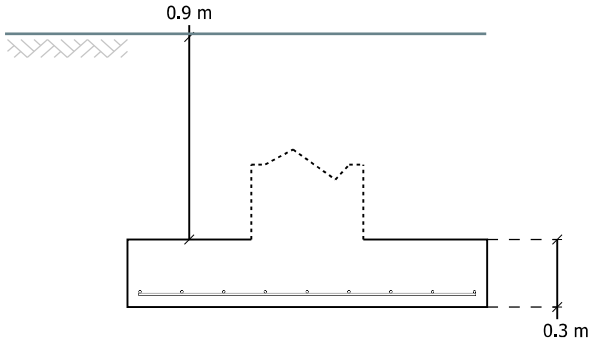
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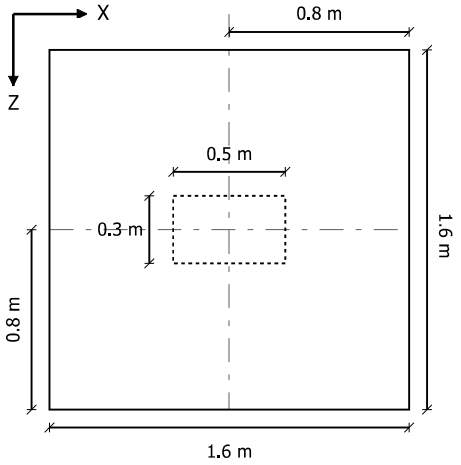
FOUNDATION-F1 ALONG GRID 1 & B

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
24	1	1.600m	1.600m	0.300m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
24	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 300.000mm
Footing Length - X (Fl) : 1600.000mm
Footing Width - Z (Fw) : 1600.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.300m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 150.000mm
Maximum Bar Spacing : 200.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 900.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.600m

Initial Width (W_o) = 1.600m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	218.877	-3.795	0.349	-0.780	0.937
9	217.395	-3.697	0.331	-1.030	0.902
10	226.494	-3.657	0.388	-0.910	0.904
11	224.589	-3.692	0.378	-0.877	0.912
12	218.866	-3.772	3.318	7.670	0.904
13	219.155	0.666	0.349	-0.780	-5.374
14	224.869	3.431	-21.730	-39.118	-7.046
15	224.501	-3.608	24.231	44.021	0.784
16	131.699	7.220	-29.268	-51.455	-10.048
17	131.209	-2.165	32.013	59.396	0.392
18	131.605	2.184	0.209	-0.468	-5.749
19	131.316	-2.254	3.178	7.982	0.529
20	219.249	5.702	-29.128	-51.767	-9.673
21	218.759	-3.683	32.152	59.084	0.767

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
15	224.501	-3.608	24.231	44.021	0.784

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
21	218.759	-3.683	32.152	59.084	0.767
22	306.427	-5.313	0.489	-1.092	1.312
23	308.161	-5.106	0.483	-1.506	1.246
24	308.161	-5.106	0.483	-1.506	1.246
25	275.137	3.264	-23.100	-41.933	-7.417
26	266.273	-4.306	51.324	94.782	0.835
27	267.057	10.710	-46.725	-82.580	-15.869
28	266.273	-4.306	51.324	94.782	0.835
29	263.050	1.819	0.419	-0.936	-7.892
30	262.637	-4.521	4.660	11.136	1.077
31	197.586	11.779	-46.850	-82.282	-16.133
32	196.801	-3.236	51.199	95.081	0.570
33	196.974	-3.382	4.555	11.370	0.795
34	197.387	2.957	0.314	-0.702	-8.173

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.560m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.470m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.600 m Governing Load Case : # 8

Width (W_2) = 1.600 m Governing Load Case : # 8

Depth (D_2) = 0.300 m Governing Load Case : # 26

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.560 m^2

Final Soil Height = 0.900 m

Footing Self Weight = 18.431 kN

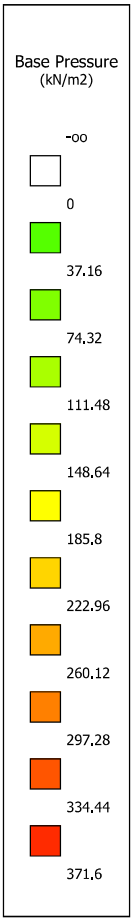
Gross Soil Bearing Capacity = 371.60 kN/m2

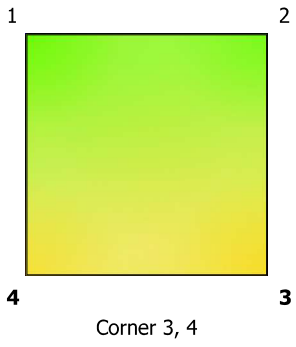
Soil Weight On Top Of Footing = 39.041 kN

Pressures at 4 Corners

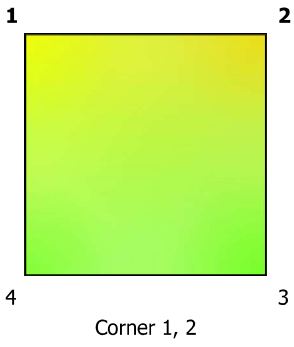
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



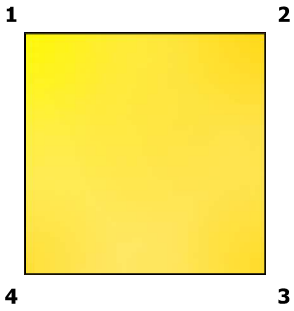


Corner 3, 4



Corner 1, 2

Governing Base Pressure Contour



Corner 1, 2, 3, 4

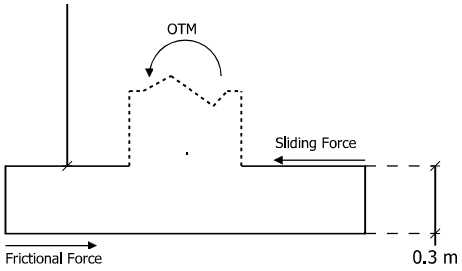
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
20	**180.0518**	213.4023	36.1369	2.7863	0.000
20	180.0518	**213.4023**	36.1369	2.7863	0.000
21	9.9643	4.4814	**205.8415**	211.3244	0.000
21	9.9643	4.4814	205.8415	**211.3244**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
20	**180.0518**	213.4023	36.1369	2.7863
20	180.0518	**213.4023**	36.1369	2.7863
21	9.9643	4.4814	**205.8415**	211.3244
21	9.9643	4.4814	205.8415	**211.3244**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	42.233	459.300	59.475	1.500	329.397	106.658	1.500
9	43.126	481.319	60.747	1.500	237.841	109.520	1.500
10	45.033	424.269	63.330	1.500	288.050	113.703	1.500
11	44.313	432.346	62.341	1.500	297.188	111.892	1.500
12	42.492	48.309	45.122	1.500	25.668	108.785	1.500
13	240.904	459.768	301.774	1.500	329.723	39.759	1.500
14	47.732	7.536	10.527	1.500	4.979	28.012	1.500
15	45.333	6.749	9.441	1.500	4.425	121.042	1.500
16	15.197	3.749	5.147	1.500	2.535	12.417	1.500
17	50.549	3.418	4.823	1.500	2.207	145.286	1.500
18	50.211	523.761	70.684	1.500	376.673	23.668	1.500
19	48.582	34.453	39.744	1.500	17.054	125.623	1.500
20	28.150	5.510	7.647	1.500	3.681	19.475	1.500
21	43.501	4.983	7.001	1.500	3.235	118.252	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 16
Governing Disturbing Force : 7.220kN
Governing Restoring Force : 109.719kN
Minimum Sliding Ratio for the Critical Load Case : 15.197
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 69.000kNm
Governing Resisting Moment : 152.303kNm
Minimum Overturning Ratio for the Critical Load Case : 2.207

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17
Governing Disturbing Force : 32.013kN
Governing Restoring Force : 109.435kN
Minimum Sliding Ratio for the Critical Load Case : 3.418
Critical Load Case for Overturning about Z-Direction : 16
Governing Overturning Moment : -12.214kNm
Governing Resisting Moment : 151.658kNm
Minimum Overturning Ratio for the Critical Load Case : 12.417

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 32.086kN

Governing Restoring Force : 154.764kN

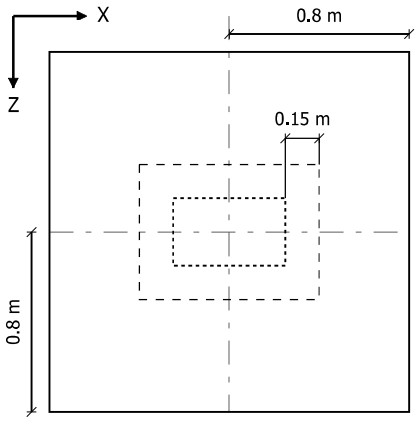
Minimum Sliding Ratio for the Critical Load Case : 4.823

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.300m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.244m

For rectangular column, β_c = B_{col} / D_{col} = 1.667

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 316.027kN, Load Case # 26

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.576m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta_c}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1358.570kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 1787.390kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 1235.063kN

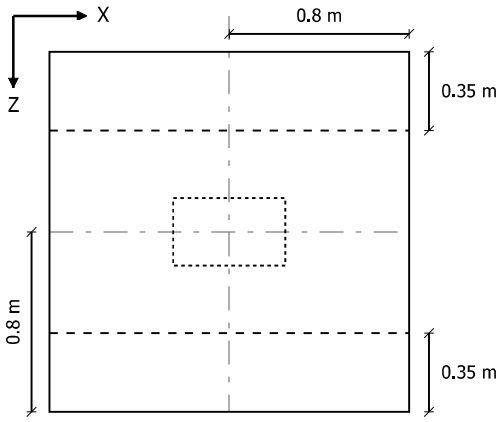
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 926.298kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 383.560kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.406m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

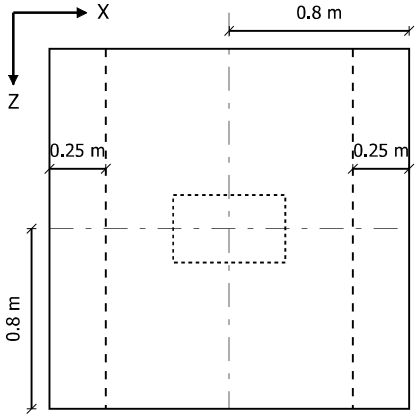
From above calculations, 0.75 X V_c = 287.670 kN

Critical load case for V_{ux} is # 26 $V_{ux} = V_{ux}|_{x=D_x}$ = 155.580 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 383.560 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.306 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

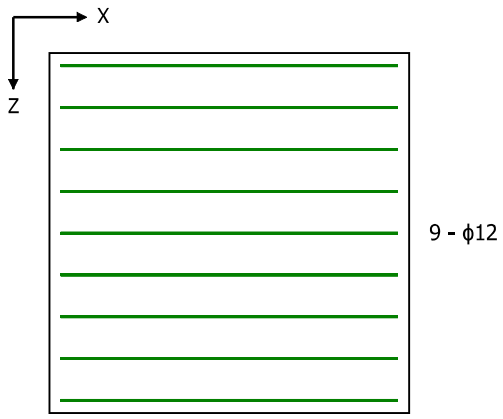
From above calculations, 0.75 X V_c = 287.670 kN

Critical load case for V_{uz} is # 23 $V_{uz} = V_{uz}|_{z=D_z}$ = 71.538 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 27

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.244	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00157	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.050	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	37.443	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	41.603	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00105	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00085	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	753.008	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 150.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover} =$ 0.500 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}} =$ 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.244 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00261

From ACI Cl.7.6.1, minimum req'd clear distance between bars

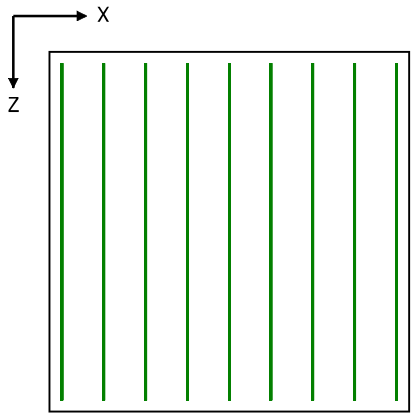
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.352

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 26

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.232 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00165

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z =	$0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ =	0.950	m
Ultimate moment,	$M_u\big\vert_{x=D_x}$ =	86.860	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	96.512	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00272	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00210	
Since	$\rho_{min} < \rho < \rho_{max}$	OK	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1009.883	mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 150.000mm

Selected spacing (S) = 186.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 185mm o.c.

Required development length for bars =	$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =	=0.410	m
Available development length for bars, D_L =	$0.5 \times \left(L - D_{col}\right) - C_{cover}$ =	0.600	m
Try bar size # 12	Area of one bar =	113.097	mm2
Number of bars required, N_{bar} =	$\frac{A_s}{A_{bar}}$ =	9	

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

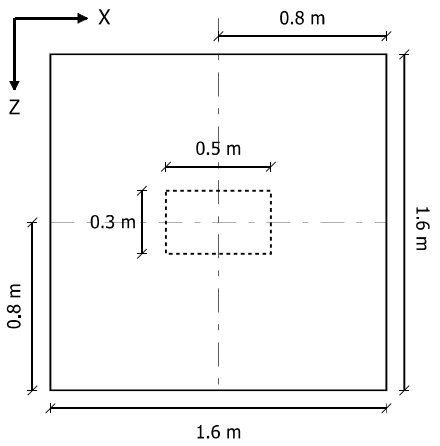
Total reinforcement area, A_{s_total} =	N_{bar} X (Area of one bar) =	1017.869	mm2
d_{eff} =	D - C_{cover} - 1.5 X (dia. of one bar) =	0.232	m
Reinforcement ratio, ρ =	$\frac{A_{s_total}}{\left(d_{eff} \times W\right)}$ =	0.00274	

From ACI Cl.7.6.1, minimum req'd clear distance between bars

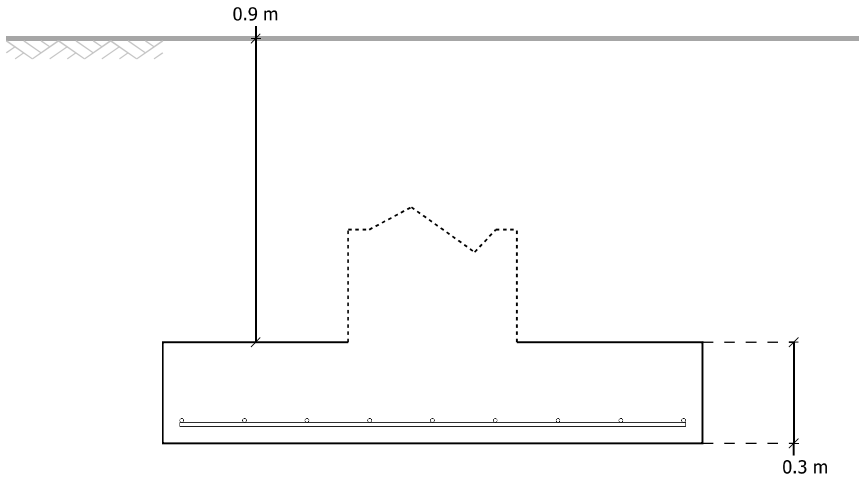
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.008

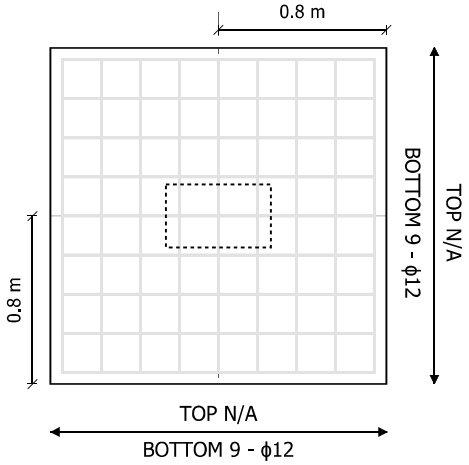
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

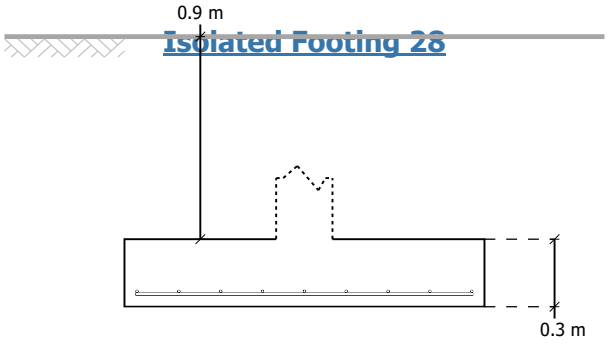
	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.6 m	1.6 m	0.3 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

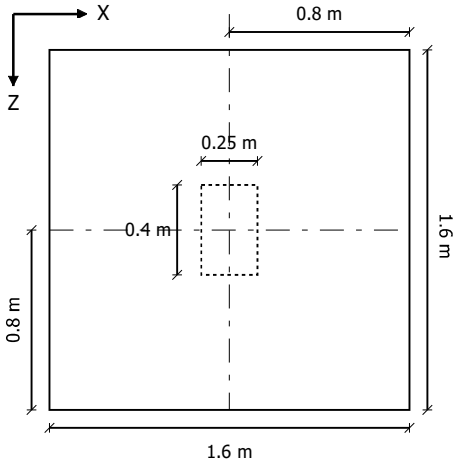
Isolated Footing Design(ACI 318-08)

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
28	1	1.600m	1.600m	0.300m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
28	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 300.000mm
Footing Length - X (Fl) : 1600.000mm
Footing Width - Z (Fw) : 1600.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.250m

Column Width - Z (B_{col}) : 0.400m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 150.000mm
Maximum Bar Spacing : 200.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 900.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.600m

Initial Width (W_o) = 1.600m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	123.947	19.057	12.853	4.340	-7.221
9	124.655	17.971	11.803	3.995	-6.874
10	127.090	18.711	12.971	4.376	-7.119
11	126.304	18.797	12.942	4.367	-7.145
12	128.287	19.047	15.631	8.571	-7.215
13	125.658	21.058	12.853	4.340	-8.788
14	127.952	21.462	11.874	4.102	-9.122
15	142.335	19.079	27.302	22.807	-7.342
16	76.566	14.987	6.289	2.250	-6.969
17	95.742	11.810	26.859	27.191	-4.596
18	76.079	13.435	7.712	2.604	-5.900
19	78.708	11.424	10.489	6.835	-4.327
20	126.144	22.610	11.430	3.986	-9.857
21	145.321	19.433	32.001	28.926	-7.484

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
15	142.335	19.079	27.302	22.807	-7.342

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
21	145.321	19.433	32.001	28.926	-7.484
22	173.526	26.680	17.995	6.075	-10.109
23	176.088	24.987	16.583	5.611	-9.572
24	176.088	24.987	16.583	5.611	-9.572
25	155.523	25.157	14.474	4.983	-10.612
26	184.506	23.297	46.119	44.565	-9.035
27	153.824	28.380	13.206	4.660	-12.832
28	184.506	23.297	46.119	44.565	-9.035
29	151.180	25.727	15.424	5.207	-10.904
30	154.936	22.854	19.392	11.252	-8.657
31	115.068	22.836	9.291	3.340	-10.717
32	145.751	17.753	42.204	43.245	-6.920
33	117.752	17.137	15.536	9.950	-6.491
34	113.996	20.010	11.568	3.906	-8.738

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.560 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.071 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.600 m Governing Load Case : # 8

Width (W_2) = 1.600 m Governing Load Case : # 8

Depth (D_2) = 0.300 m Governing Load Case : # 28

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.560 m^2

Final Soil Height = 0.900 m

Footing Self Weight = 18.431 kN

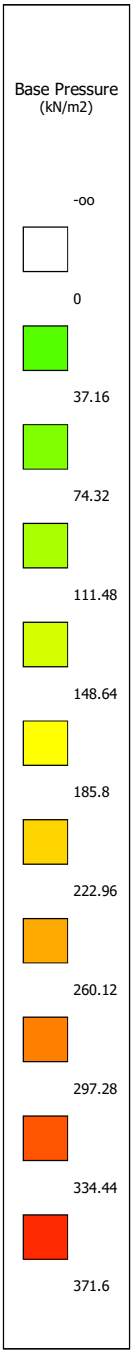
Gross Soil Bearing Capacity = 371.60 kN/m2

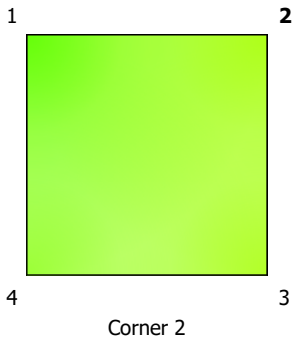
Soil Weight On Top Of Footing = 39.851 kN

Pressures at 4 Corners

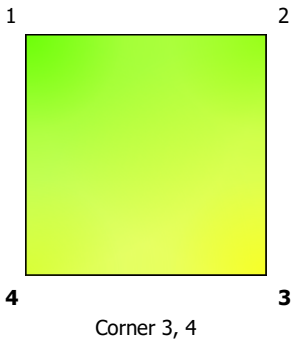
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

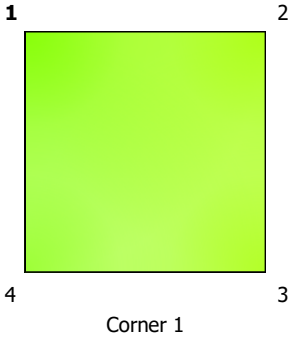




Corner 2

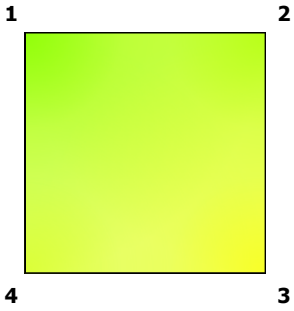


Corner 3, 4



Corner 1

Governing Base Pressure Contour



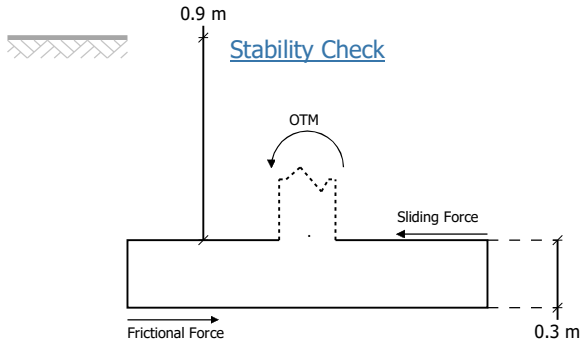
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
9	**42.4546**	78.3873	100.4646	64.5319	0.000
20	36.8038	**85.5550**	107.2793	58.5282	0.000
21	3.5934	42.5988	**155.4714**	116.4660	0.000
21	3.5934	42.5988	155.4714	**116.4660**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
9	**42.4546**	78.3873	100.4646	64.5319
20	36.8038	**85.5550**	107.2793	58.5282
21	3.5934	42.5988	**155.4714**	116.4660
21	3.5934	42.5988	155.4714	**116.4660**



Stability Check

-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	5.546	8.223	6.503	1.500	17.819	11.337	1.500
9	5.904	8.990	6.979	1.500	19.455	12.005	1.500
10	5.746	8.289	6.679	1.500	17.970	11.717	1.500
11	5.696	8.272	6.634	1.500	17.932	11.621	1.500
12	5.681	6.923	6.211	1.500	11.275	11.613	1.500
13	5.066	8.300	6.116	1.500	17.986	9.801	1.500
14	5.033	9.097	6.228	1.500	19.474	9.632	1.500
15	6.099	4.262	4.940	1.500	5.186	12.352	1.500
16	5.219	12.437	6.805	1.500	26.139	9.487	1.500
17	7.564	3.326	4.306	1.500	3.503	15.249	1.500
18	5.800	10.105	7.114	1.500	21.912	10.914	1.500
19	6.955	7.575	7.245	1.500	11.005	14.248	1.500
20	4.731	9.358	5.971	1.500	19.932	8.920	1.500
21	6.077	3.690	4.461	1.500	4.234	12.301	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20
Governing Disturbing Force : 22.610kN
Governing Restoring Force : 106.967kN
Minimum Sliding Ratio for the Critical Load Case : 4.731
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 35.248kNm
Governing Resisting Moment : 123.476kNm
Minimum Overturning Ratio for the Critical Load Case : 3.503

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17

Governing Disturbing Force : 26.859kN

Governing Restoring Force : 89.334kN

Minimum Sliding Ratio for the Critical Load Case : 3.326

Critical Load Case for Overturning about Z-Direction : 20

Governing Overturning Moment : -16.640kNm

Governing Resisting Moment : 148.429kNm

Minimum Overturning Ratio for the Critical Load Case : 8.920

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 29.341kN

Governing Restoring Force : 126.337kN

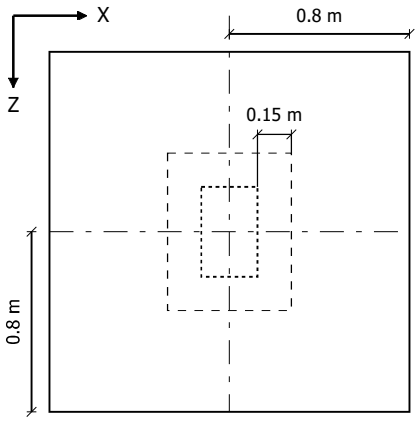
Minimum Sliding Ratio for the Critical Load Case : 4.306

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.300m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.244m

For rectangular column, β_c = B_{col} / D_{col} = 1.600

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 220.658kN, Load Case # 28

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.276m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1227.632kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 1715.472kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 1091.228kN

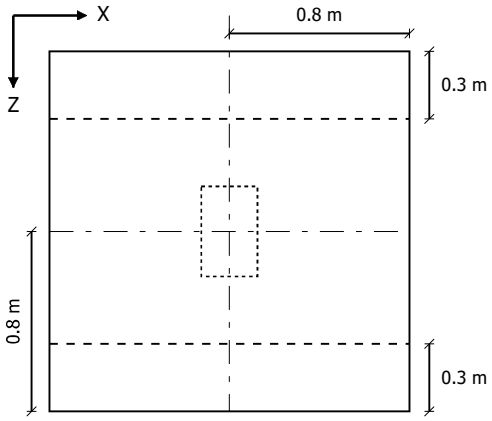
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 818.421kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 383.560kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.356m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

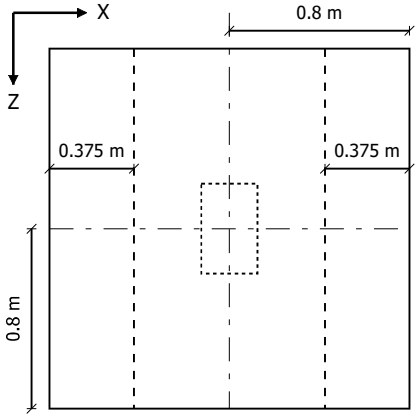
From above calculations, 0.75 X V_c = 287.670 kN

Critical load case for V_{ux} is # 28 $V_{ux} = V_{ux}|_{x=D_x}$ = 91.762 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 383.560 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.431 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

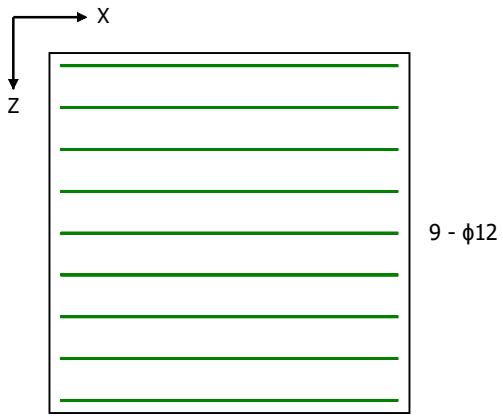
From above calculations, 0.75 X V_c = 287.670 kN

Critical load case for V_{uz} is # 26 $V_{uz} = V_{uz}|_{z=D_z}$ = 77.085 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.244	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00160	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.925	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	40.838	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	45.376	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00114	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00093	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	769.450	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 150.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars = =0.410 m

$$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.625 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.244 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00261

From ACI Cl.7.6.1, minimum req'd clear distance between bars

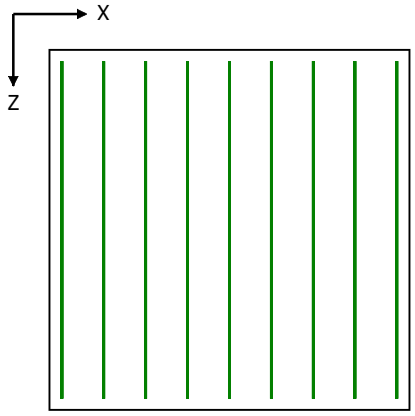
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.323

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.232 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00157

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.000 m

Ultimate moment, $M_u|_{x=D_x}$ = 45.957 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 51.063 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00143

(Based on gross depth) ρ x d_{eff} / Depth = 0.00110

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 755.524 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 150.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.550 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.232 m

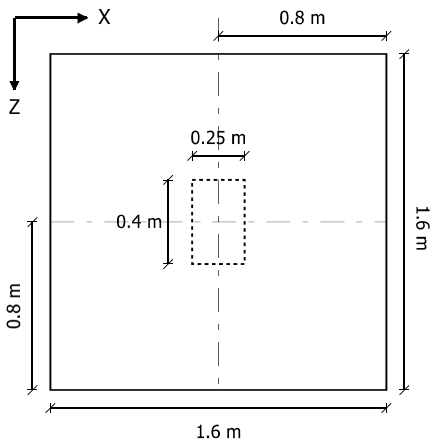
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00274

From ACI Cl.7.6.1, minimum req'd clear distance between bars

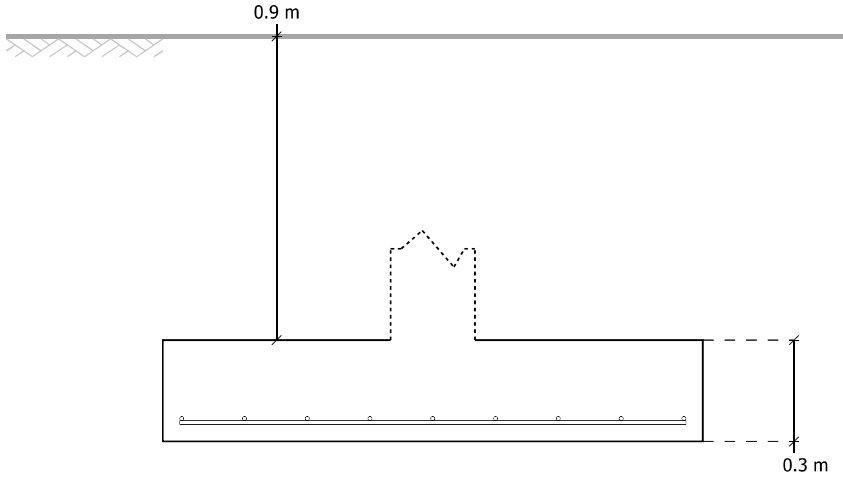
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.347

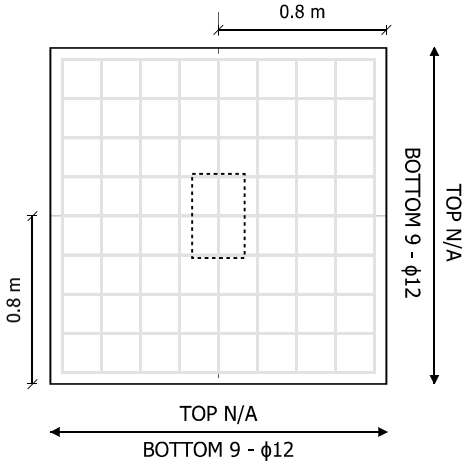
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.6 m	1.6 m	0.3 m	N/A	N/A	9 - ϕ12	9 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

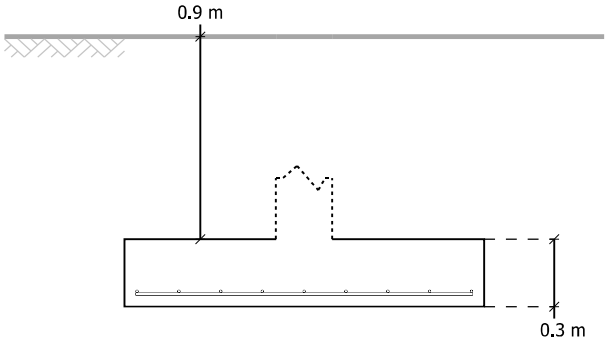
Print Calculation Sheet

file:///C:/Staad%20Foundation%20Advanced%20SS3/CalcXsl/footing.xml

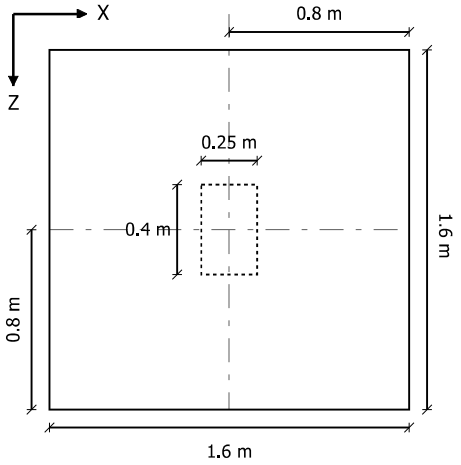
FOUNDATION-F1 ALONG GRID 2 & C

Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
28	1	1.600m	1.600m	0.300m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
28	9 - ϕ12	9 - ϕ12	N/A	N/A	N/A	N/A



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 300.000mm
Footing Length - X (Fl) : 1600.000mm
Footing Width - Z (Fw) : 1600.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.250m

Column Width - Z (B_{col}) : 0.400m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 150.000mm
Maximum Bar Spacing : 200.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 900.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.580
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.600m

Initial Width (W_o) = 1.600m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
8	SLS: (DL)	1.00	1.00	1.00
9	SLS: (DL+T)	1.00	1.00	1.00
10	SLS: (D+LR)	1.00	1.00	1.00
11	SLS: (D+0.75LR)	1.00	1.00	1.00
12	SLS: (D+0.7EZ)	1.00	1.00	1.00
13	SLS: (D+0.7EX)	1.00	1.00	1.00
14	SLS: (D+0.75WX+0.75LR)	1.00	1.00	1.00
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
16	SLS: (0.6D+WX)	1.00	1.00	1.00
17	SLS: (0.6D+WZ)	1.00	1.00	1.00
18	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
19	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
20	SLS: (D+WX)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
15	SLS: (D+0.75WZ+0.75LR)	1.00	1.00	1.00
21	SLS: (D+WZ)	1.00	1.00	1.00
22	ULS: (1.4 DL)	1.00	1.00	1.00
23	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
24	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
25	ULS: (1.2D+0.80WX+1.6LR)	1.00	1.00	1.00
26	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+1.6WX+0.5LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+EX)	1.00	1.00	1.00
30	ULS: (1.2D+EZ)	1.00	1.00	1.00
31	ULS: (0.90D+1.6WX)	1.00	1.00	1.00
32	ULS: (0.90D+1.6WZ)	1.00	1.00	1.00
33	ULS: (0.90D+EZ)	1.00	1.00	1.00
34	ULS: (0.90D+EX)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
8	123.947	19.057	12.853	4.340	-7.221
9	124.655	17.971	11.803	3.995	-6.874
10	127.090	18.711	12.971	4.376	-7.119
11	126.304	18.797	12.942	4.367	-7.145
12	128.287	19.047	15.631	8.571	-7.215
13	125.658	21.058	12.853	4.340	-8.788
14	127.952	21.462	11.874	4.102	-9.122
15	142.335	19.079	27.302	22.807	-7.342
16	76.566	14.987	6.289	2.250	-6.969
17	95.742	11.810	26.859	27.191	-4.596
18	76.079	13.435	7.712	2.604	-5.900
19	78.708	11.424	10.489	6.835	-4.327
20	126.144	22.610	11.430	3.986	-9.857
21	145.321	19.433	32.001	28.926	-7.484

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
15	142.335	19.079	27.302	22.807	-7.342

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
21	145.321	19.433	32.001	28.926	-7.484
22	173.526	26.680	17.995	6.075	-10.109
23	176.088	24.987	16.583	5.611	-9.572
24	176.088	24.987	16.583	5.611	-9.572
25	155.523	25.157	14.474	4.983	-10.612
26	184.506	23.297	46.119	44.565	-9.035
27	153.824	28.380	13.206	4.660	-12.832
28	184.506	23.297	46.119	44.565	-9.035
29	151.180	25.727	15.424	5.207	-10.904
30	154.936	22.854	19.392	11.252	-8.657
31	115.068	22.836	9.291	3.340	-10.717
32	145.751	17.753	42.204	43.245	-6.920
33	117.752	17.137	15.536	9.950	-6.491
34	113.996	20.010	11.568	3.906	-8.738

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 2.560 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.071 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.600 m Governing Load Case : # 8

Width (W_2) = 1.600 m Governing Load Case : # 8

Depth (D_2) = 0.300 m Governing Load Case : # 28

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 2.560 m^2

Final Soil Height = 0.900 m

Footing Self Weight = 18.431 kN

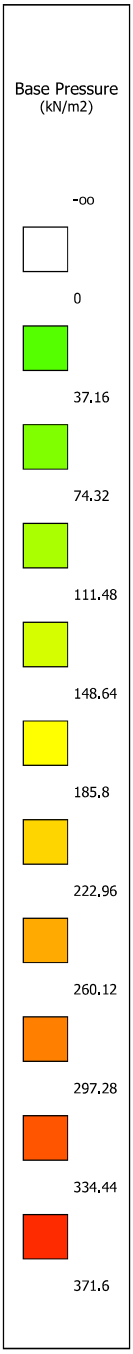
Gross Soil Bearing Capacity = 371.60 kN/m2

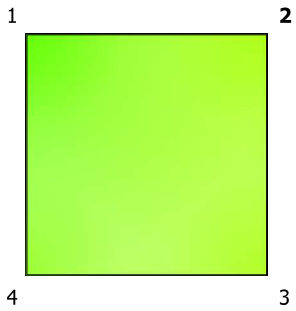
Soil Weight On Top Of Footing = 39.851 kN

Pressures at 4 Corners

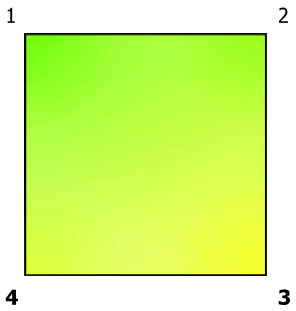
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

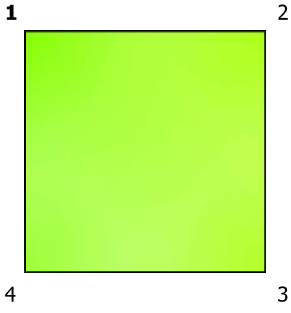




Corner 2

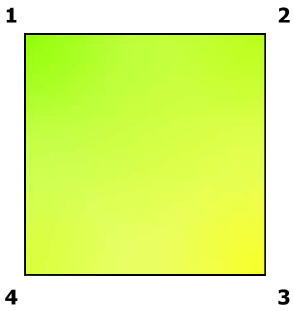


Corner 3, 4



Corner 1

Governing Base Pressure Contour



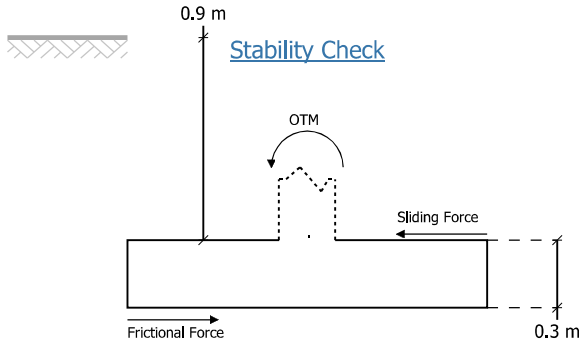
Corner 1, 2, 3, 4

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
9	**42.4546**	78.3873	100.4646	64.5319	0.000
20	36.8038	**85.5550**	107.2793	58.5282	0.000
21	3.5934	42.5988	**155.4714**	116.4660	0.000
21	3.5934	42.5988	155.4714	**116.4660**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
9	**42.4546**	78.3873	100.4646	64.5319
20	36.8038	**85.5550**	107.2793	58.5282
21	3.5934	42.5988	**155.4714**	116.4660
21	3.5934	42.5988	155.4714	**116.4660**



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
8	5.546	8.223	6.503	1.500	17.819	11.337	1.500
9	5.904	8.990	6.979	1.500	19.455	12.005	1.500
10	5.746	8.289	6.679	1.500	17.970	11.717	1.500
11	5.696	8.272	6.634	1.500	17.932	11.621	1.500
12	5.681	6.923	6.211	1.500	11.275	11.613	1.500
13	5.066	8.300	6.116	1.500	17.986	9.801	1.500
14	5.033	9.097	6.228	1.500	19.474	9.632	1.500
15	6.099	4.262	4.940	1.500	5.186	12.352	1.500
16	5.219	12.437	6.805	1.500	26.139	9.487	1.500
17	7.564	3.326	4.306	1.500	3.503	15.249	1.500
18	5.800	10.105	7.114	1.500	21.912	10.914	1.500
19	6.955	7.575	7.245	1.500	11.005	14.248	1.500
20	4.731	9.358	5.971	1.500	19.932	8.920	1.500
21	6.077	3.690	4.461	1.500	4.234	12.301	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 20
Governing Disturbing Force : 22.610kN
Governing Restoring Force : 106.967kN
Minimum Sliding Ratio for the Critical Load Case : 4.731
Critical Load Case for Overturning about X-Direction : 17
Governing Overturning Moment : 35.248kNm
Governing Resisting Moment : 123.476kNm
Minimum Overturning Ratio for the Critical Load Case : 3.503

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 17

Governing Disturbing Force : 26.859kN

Governing Restoring Force : 89.334kN

Minimum Sliding Ratio for the Critical Load Case : 3.326

Critical Load Case for Overturning about Z-Direction : 20

Governing Overturning Moment : -16.640kNm

Governing Resisting Moment : 148.429kNm

Minimum Overturning Ratio for the Critical Load Case : 8.920

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 17

Governing Disturbing Force : 29.341kN

Governing Restoring Force : 126.337kN

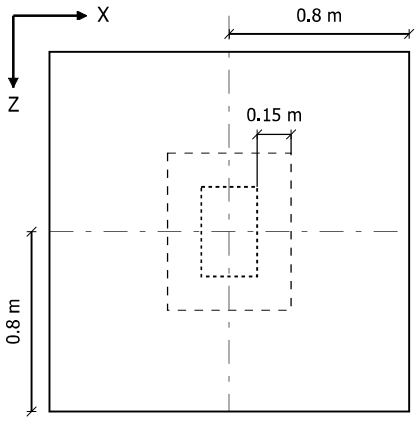
Minimum Sliding Ratio for the Critical Load Case : 4.306

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.300m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.244m

For rectangular column, β_c = B_{col} / D_{col} = 1.600

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 220.658kN, Load Case # 28

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.276m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1227.632kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 1715.472kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_{.c}'} \times b_{.o} \times d_{.eff}$ 1091.228kN

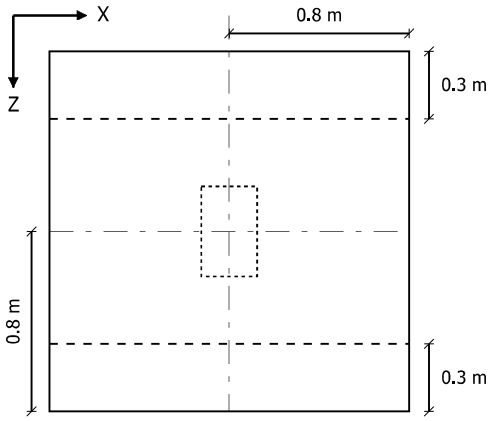
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 818.421kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 383.560kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{zd}$ = 0.356m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

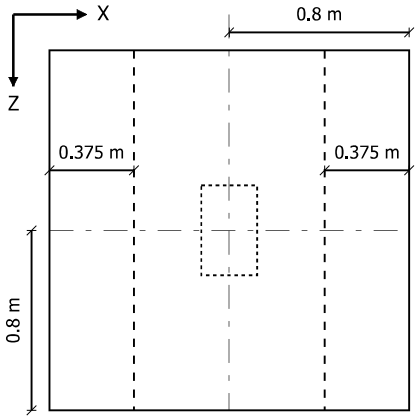
From above calculations, 0.75 X V_c = 287.670 kN

Critical load case for V_{ux} is # 28 $V_{ux} = V_{ux}|_{x=D_x}$ = 91.762 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 383.560 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.431 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

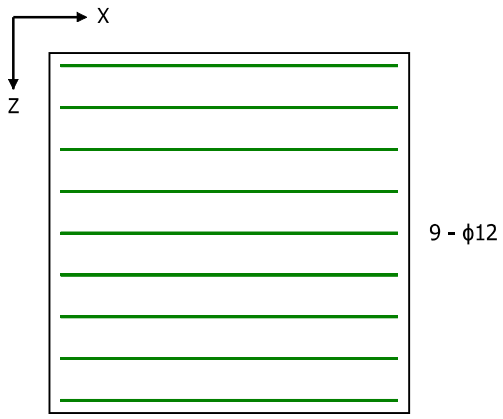
From above calculations, 0.75 X V_c = 287.670 kN

Critical load case for V_{uz} is # 26 $V_{uz} = V_{uz}|_{z=D_z}$ = 77.085 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.244	m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796	
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317	
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488	
From ACI Cl. 7.12.2, ρ_{min} =		0.00160	
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118	

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	0.925	m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	40.838	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	45.376	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00114	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00093	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	769.450	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 150.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}} =$

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover} =$ 0.625 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}} =$ 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.244 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)} =$ 0.00261

From ACI Cl.7.6.1, minimum req'd clear distance between bars

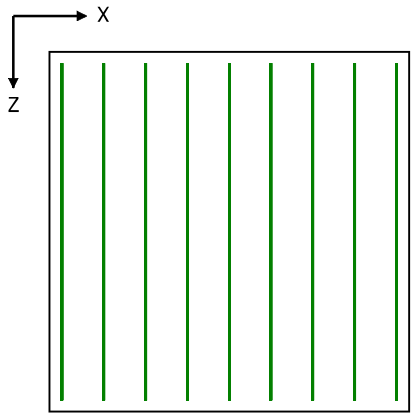
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.323

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



9 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.232 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal} =$ 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00157

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')} =$ 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.000 m

Ultimate moment, $M_u|_{x=D_x}$ = 45.957 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 51.063 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00143

(Based on gross depth) ρ x d_{eff} / Depth = 0.00110

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 755.524 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 150.000mm

Selected spacing (S) = 200.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 200mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.550 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 9

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1017.869 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.232 m

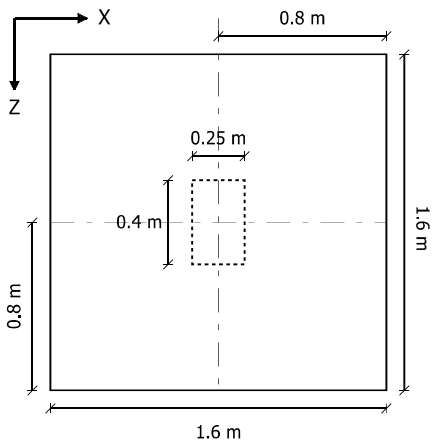
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00274

From ACI Cl.7.6.1, minimum req'd clear distance between bars

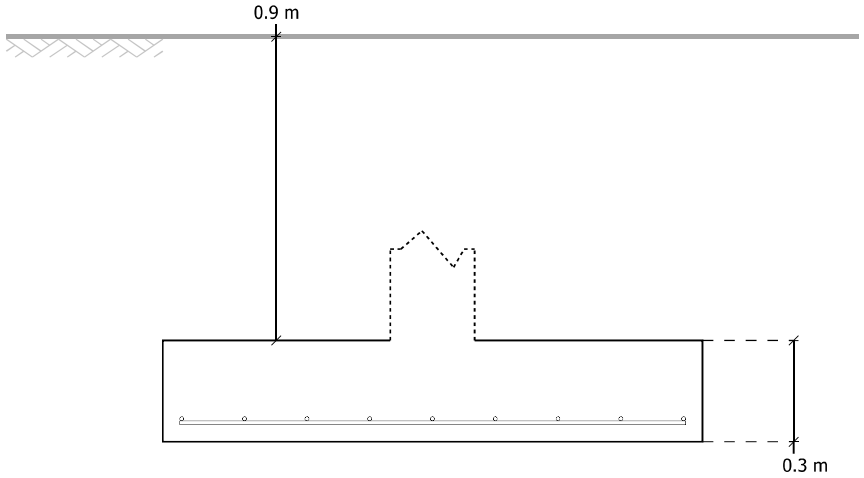
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 150.000mm

Provided Steel Area / Required Steel Area = 1.347

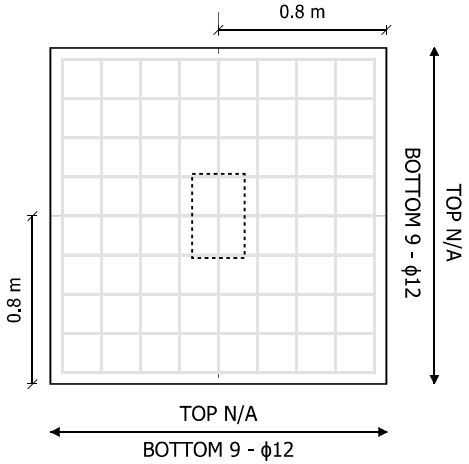
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
Foundation	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
	1.6 m	1.6 m	0.3 m	N/A	N/A	9 - ϕ12	9 - ϕ12
Pedestal	l	b	h	Main Reinforcement		Trans Reinforcement	
	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet

file:///C:/Staad%20Foundation%20Advanced%20SS3/CalcXsl/footing.xml

ATTACHMENT 3 COMMENT RESOLUTION LOG SHEET

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT** **PROJECT NO.: 216**
STAGE:	**4 - PROJECT EXECUTION.** **REVIEW LOG DOC. NO. & REV.:**
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-001622, 12 March 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 12 March 2020	Comment by	EPC Response DATE: 26 March,2020	Status	Remarks
			General Comments (cc)				
MD-216-6000-EG-CV-CAL-1489-A01	6000	1	Use the newer versions of Saudi Building Codes	Mr.Klaus Grof (Bechtel)	Codes & relevant load combinations were followed as per Section-4.1.1 from MD-216-1000-EG-ST-CR-001-A02. Also as Basic wind speed is provided in Design Criteria, load combinations from SBC-2007 equivalent to ASCE-2005 were considered in Analysis & Design.	CLOSED	